

The WKB Geometry of Newton Polygons: A Functorial Classification Theory

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Abstract

For a scalar local linear operator L (differential, difference, or q -difference) with one irregular singularity at $x = \infty$ whose Newton polygon has a unique dominant edge of slope $-p/q$ with $r+1$ lattice points, the WKB ansatz on that edge produces a characteristic polynomial $\chi \in \mathbb{C}[c]$. Our central algebraic identity is

$$\chi(c) = \chi_w(c^q),$$

where $\chi_w \in \mathbb{C}[w]$ is the classical edge-indicial polynomial of degree r in $w = c^q$. The exponent constellation $\mathcal{C}(L) \subset \mathbb{C}$ is therefore a μ_q -equivariant union of r fibres of $c \mapsto c^q$ over $\text{Roots}(\chi_w)$.

This factorisation organises a finite stratified atlas: a computational sweep yields 23 single-edge classes for $-p/q \in \{1/2, 1/3, 1/4, 2/3, 3/2\}$ and $r \leq 3$, refining to 39 once multi-edge, nested-ramification, and prime- q arithmetic strata are tracked. We prove a per-edge factorisation along the slope filtration, an iterated $\mu_{q'}$ -pullback law for nested inner polynomials, a block-diagonal pullback for systems, and a prime- q Kummer dichotomy separating the algebraically independent stratum from the cyclotomic stratum.

The analytic input attaching anti-Stokes directions is classical (Levelt–Turrittin, Sibuya, Wasow); on the q -fold cover the rays of a generic single-edge operator satisfy $\arg \Delta_{ij} + p\phi \equiv \pi/2 \pmod{\pi}$. One explicit consequence: the slope numerator p , invisible to $\mathcal{C}(L)$, is visible to the cover-side ray pattern. The analytic statements (Theorems 7.1, 7.4, 7.6, 7.7) are $\mu_q/\mu_{q'}$ -equivariant repackagings of classical formal-classification theory.

KEYWORDS: WKB analysis; Newton polygon; irregular singularity; formal classification; Levelt–Turrittin; Stokes phenomenon; anti-Stokes directions; wild character variety; functorial classification; μ_q -pullback.

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1 Introduction

1.1 The constellation map

To a scalar linear operator L with a single irregular singularity at $x = \infty$, the WKB ansatz on the dominant Newton edge attaches a finite multiset of complex numbers, the *exponent constellation*

$$\mathcal{C}(L) \subset \mathbb{C}.$$

The present paper is about the combinatorial geometry of the assignment

$$F: L \longmapsto \mathcal{C}(L),$$

which we call the *constellation map*. An earlier draft of this paper used the word “functor” for F informally; this revision reserves that word for the precise categorical setting given in Theorem 1.1 below and otherwise speaks of the *constellation map*. In the body of the paper F is treated as a map between sets of objects (scalar local models with one dominant Newton edge on the source side, finite multisets in $\mathbb{C}$ up to global rotation on the target). Theorem 1.1 promotes

this to a functor between two small categories whose morphisms are formal gauge equivalences preserving the designated edge (source) and rotation-compatible bijections of multisets (target); no statement in the paper depends on the morphism part of F , but the formulation pins down what we mean by “the fibres of F ”.

Definition 1.1 (The category $\mathfrak{L}$ and the constellation functor). Let $\mathfrak{L}$ be the category whose

- *objects* are pairs $(L, \mathfrak{e})$, where L is a scalar linear operator on $\mathbb{C}[x]$ in $\{D, S, T\}$ with exactly one irregular singularity at $x = \infty$ and $\mathfrak{e}$ is a designated dominant edge of $N(L)$ at infinity;
- *morphisms* $(L, \mathfrak{e}) \rightarrow (L', \mathfrak{e}')$ are formal gauge equivalences $\widehat{M}(L) \xrightarrow{\sim} \widehat{M}(L')$ over $\mathbb{C}((x^{-1}))$ that send $\mathfrak{e}$ to $\mathfrak{e}'$ under the induced action on Newton polygons.

Let $\mathfrak{C}$ be the category whose

- *objects* are finite multisets $C \subset \mathbb{C}$ taken up to the global rotation equivalence $C \sim e^{i\theta} C$ for $\theta \in \mathbb{R}/2\pi\mathbb{Z}$;
- *morphisms* are multiset bijections compatible with the rotation action.

The assignment $(L, \mathfrak{e}) \mapsto \mathcal{C}(L)$, extended to morphisms by sending a formal gauge equivalence to the induced bijection of roots of χ , defines a functor

$$F: \mathfrak{L} \longrightarrow \mathfrak{C}.$$

Remark 1.2. Functoriality is immediate from the construction of χ from the leading-coefficient data along $\mathfrak{e}$: a formal gauge equivalence preserving $\mathfrak{e}$ acts on the LC_∞ -data of $\mathfrak{e}$ by a cocycle that preserves χ up to an overall rescaling of c , and hence preserves the multiset of roots of χ up to a global rotation. We will use “ F ” and “the constellation map” interchangeably in the remainder of the paper. Whenever only the underlying object map matters, the reader may forget Theorem 1.1 without loss.

1.2 The structural law

The central observation, proved in Theorem 3.1, is the algebraic identity

$$\chi(c) = \chi_w(c^q),$$

where χ is the WKB characteristic polynomial built from the LC_∞ -data along the dominant edge and $\chi_w \in \mathbb{C}[w]$ is the *inner polynomial* of degree r . Solving $\chi = 0$ amounts to solving χ_w in the variable $w = c^q$ and pulling back along $c \mapsto c^q$; the constellation decomposes canonically as a μ_q -equivariant union of r fibres over $\text{Roots}(\chi_w)$. This factorisation is at once an obstruction (no multiset outside the image of F admits it) and a recipe (every multiset in the image is so built). Section 3 states the law as Theorem 3.1 with two corollaries (ring decomposition; Vieta monodromy) and a remark on the degeneracy strata.

1.3 χ_w as the edge-indicial polynomial

The polynomial χ_w is not new content. It is exactly the *edge-indicial polynomial* of the dominant edge $\mathfrak{e}$ in the variable $w = c^q$, the form that appears in the classical leading-balance analysis of irregular singularities (Wasow [Was87], Sibuya [Sib90], Sabbah [Sab07]). Setting $w = c^q$, the equation $\chi_w(w) = 0$ is the standard requirement that the WKB ansatz produce a non-trivial formal solution along $\mathfrak{e}$. The novelty of Theorem 3.1 is therefore not the existence of χ_w but the clean identification

$$\chi(c) = \chi_w(c^q),$$

together with the consequence that the μ_q -equivariant fibre structure of $\mathcal{C}(L)$ becomes explicit, and the slope denominator q appears as a *pullback index* rather than as a feature of the leading balance to be rediscovered case by case.

1.4 From a law to a classification

The structural law turns the classification problem into a combinatorial one. Up to global rotation, a constellation is recovered from (p, q, r) together with the modulus-with-multiplicity partition π of the roots of χ_w . A computational sweep, recorded in the companion dataset [Pap26], enumerates these classes for slopes $-p/q \in \{1/2, 1/3, 1/4, 2/3, 3/2\}$ and ranks $r \in \{1, 2, 3\}$ across three equation types: there are exactly 23 classes in the single-edge case. Once multi-edge polygons, nested ramification, and the prime- q arithmetic stratification are recorded, the atlas refines to 39 classes. These populate the open stratum and four named degenerate strata (equal modulus; resonant multiplicity; zero root; accidental symmetry); Section 8 catalogues them.

The classification is extended in several directions:

- beyond the single edge: per-edge factorisation along the slope filtration (Theorem 4.3, Section 4);
- beyond a single μ_q -orbit: nested ramification with iterated pullbacks (Theorem 4.6, Section 4);
- beyond squarefree χ_w : degeneracy strata and a prime- q Kummer dichotomy (Theorem 5.1, Section 5);
- beyond scalar operators: block-diagonal systems and a symmetric-power target (Theorem 6.1, Section 6);
- beyond the discrete invariant: an analytic refinement attaching anti-Stokes directions to the differences $\Delta_{ij} = c_i - c_j$ (Theorems 7.1, 7.4, 7.6 and 7.7, Section 7).

1.5 Scope of the analytic refinement

The analytic theorems of Section 7 attach Stokes-side information to the constellation. The reader should bear in mind a clear division of labour:

- *Classical content.* The Levelt–Turrittin slope decomposition of a formal connection on a ramified cover of index q ; the existence of formal exponential factors $Q_i(t) = c_i t^p + (\text{lower})$; the principle that anti-Stokes rays are governed by the leading term of $Q_i - Q_j$; and the slope-graded structure of Stokes data for multi-edge connections are classical (Sibuya [Sib90], Wasow [Was87], Sabbah [Sab05]).
- *Algebraic refinement.* What is new in the present paper is the explicit linear form for the anti-Stokes condition in Theorem 7.1,

$$\arg \Delta_{ij} + p\phi \equiv \frac{\pi}{2} \pmod{\pi};$$

the observation that nested factorisations of χ_w induce an additional $\mu_{q'}$ -action on the discrete part of the wild data *without* changing the ramification index q (Theorem 7.4); the consequent $\mu_q/\mu_{q'}$ symmetry bookkeeping; and the p -visibility statement Theorem 7.7, by which the slope numerator p —invisible to $\mathcal{C}(L)$ —is visible to the cover-side ray pattern.

- *Equivariance, not Stokes matrices.* Theorem 7.4 asserts $\mu_{q'}$ -equivariance of an irregular-type parametrisation; it is not a statement about wild character variety strata in the moduli-theoretic sense of Boalch [Boa14, Boa16]. We are explicit about this distinction.

1.6 What is and is not classified

The object classified in this paper is the image of F over the single-edge (23-class) and the extended single-edge / multi-edge / nested-ramification / system / arithmetic strata (39-class). We do *not* classify operators. The constellation is a coarsening of the irregular-type data of

Levelt–Turrittin, and the kernel of this coarsening—operators with identical constellation but distinct Stokes structures—is the complementary problem solved (in appropriate categories) by wild character varieties (Boalch [Boa14, Boa16]). Section 9 positions the present work alongside neighbouring programmes: ADE classifications of singularities; Stokes-graph and spectral-network classifications; the formal classification of irregular connections (Levelt–Turrittin, Sabbah [Sab07, Sab05], Mochizuki [Moc11]); and wild character varieties. The constellation classification is orthogonal to each.

1.7 Roadmap

Section 2 fixes notation, recalls Newton polygons and the WKB symbol, and includes a short notation table. Section 3 proves the structural law (Theorem 3.1). Section 4 extends to multi-edge polygons via the slope- filtration per-edge factorisation (Theorem 4.3) and to nested ramification (Theorem 4.6). Section 5 treats the four degeneracy strata and proves the prime- q Kummer dichotomy (Theorem 5.1), separating the algebraic statement (Kummer theory over $\mathbb{Q}(\zeta_q)$) from the empirical bounded-search corroboration. Section 6 treats systems (Theorem 6.1). Section 7 states and proves the four analytic-lifting theorems (Theorems 7.1, 7.4, 7.6 and 7.7), with explicit hypotheses (OS1)–(OS5) collected in Section 7.1. Section 8 presents the 23-class atlas, the 39-class refinement, the taxonomy diagram (Figure 1), and the constellation zoo (Figure 2). Section 9 positions the work alongside its neighbours, Section 10 closes with future directions, and Section 11 contains a reproducibility note.

The reader interested only in the structural law may read Section 3 immediately after Section 2; the analytic theorems of Section 7 depend logically on Sections 3 and 4 but may be read in any internal order.

2 Newton polygons and the WKB symbol

2.1 Operators and conventions

We work over $\mathbb{C}$. Let L be a scalar linear operator

$$L = \sum_{k=0}^n a_k(x) \square^k, \quad a_k \in \mathbb{C}[x], \quad a_n \neq 0,$$

where $\square$ is one of:

- $D := d/dx$ (the *ODE case*),
- $S: f(x) \mapsto f(x+1)$ (the *difference case*),
- $T: f(x) \mapsto f(qx)$ for fixed $|q| \neq 0, 1$ (the *q -difference case*; we reserve the symbol q for the slope denominator below and use a separate parameter for the q -shift when both appear).

We assume L has a single irregular singularity at $x = \infty$ and is regular elsewhere. The coefficients a_k are polynomial throughout; the extension to formal Laurent series is straightforward and notationally heavier without affecting the structural law.

2.2 Newton polygon at infinity

For each $a_k \in \mathbb{C}[x] \setminus \{0\}$, write $\deg a_k$ for its degree in x . The *Newton polygon at infinity* of L is

$$N(L) := \text{ConvexHull}\{ (k, \deg a_k) : a_k \neq 0 \} - \text{lower half-plane},$$

i.e. the upper convex hull of the support, viewed as a polygon in $\mathbb{R}^2$. Edges of $N(L)$ are line segments with slope $-p/q$ in lowest terms ($\gcd(p, q) = 1$, $p, q \geq 1$), oriented so that the slope decreases left-to-right.

Definition 2.1 (Dominant edge). A *dominant edge* of $N(L)$ is an edge of strictly maximal slope. We say L has *one dominant edge* if $N(L)$ admits a unique such edge. Multi-edge polygons are treated in §4.

Remark 2.2 (Equation-type independence of $N(L)$). The Newton polygon is defined uniformly across D, S, T: the recipe “plot $(k, \deg a_k)$ and take the upper convex hull” produces the same polygon. The *analytic* interpretation of an edge differs (slope $-p/q$ means $f \sim \exp(cx^{p/q})$ for D, but $f \sim \prod(\text{shift factors})$ for S or T); the *combinatorial* content is identical.

Lemma 2.3 (Lattice points on an edge). *Let E be an edge of $N(L)$ of slope $-p/q$ with $\gcd(p, q) = 1$. Then the lattice points of E are exactly*

$$E \cap \mathbb{Z}^2 = \{(k_0 + jq, h_0 - jp) : 0 \leq j \leq r\},$$

where (k_0, h_0) is the lower-right endpoint of E and $r \geq 1$ is the rank multiplier. Equivalently, E contains $r + 1$ lattice points and has horizontal extent rq .

Proof. A vector $(\Delta k, \Delta h)$ along E satisfies $\Delta h / \Delta k = -p/q$. The primitive integer vector in this direction is $(q, -p)$, and $\gcd(q, p) = 1$ ensures these are the only lattice points lying on E between its two endpoints, parametrised by $j = 0, 1, \dots, r$ where r is the integer length of E measured in multiples of $(q, -p)$. $\square$

2.3 The WKB symbol and the characteristic polynomial

Fix a dominant edge E of slope $-p/q$ with lattice points $(k_0 + jq, h_0 - jp)$, $0 \leq j \leq r$, as above. For each such lattice point, write

$$\alpha_j := \text{LC}_\infty(a_{k_0+jq}) \in \mathbb{C},$$

where LC_∞ extracts the leading coefficient in x at ∞ . (The lattice point lies on E iff a_{k_0+jq} has degree exactly $h_0 - jp$ in x , so α_j is unambiguous.)

Definition 2.4 (WKB characteristic polynomial). The *WKB characteristic polynomial* of L along the dominant edge E is

$$\chi(c) := \sum_{j=0}^r \alpha_j c^{(r-j)q} \in \mathbb{C}[c].$$

The *inner polynomial* is

$$\chi_w(w) := \sum_{j=0}^r \alpha_j w^{r-j} \in \mathbb{C}[w], \quad \deg \chi_w = r.$$

The *constellation* of L is the multiset $\mathcal{C}(L) := \text{Roots}(\chi) \subset \mathbb{C}$, with roots counted with multiplicity.

Remark 2.5 (Origin of χ from the WKB ansatz). For $\square = \text{D}$, substituting $f = \exp(\frac{q}{p} c x^{p/q}) g(x)$ into $Lf = 0$ with c to be determined gives, at leading order in x , the equation $\sum_j \alpha_j c^{(r-j)q} x^{(r-j)p+h_0} = 0$ along the edge E . Equating coefficients of the highest power of x yields $\chi(c) = 0$, recovering Theorem 2.4. For S, the substitution is $f \sim \exp(\frac{cx^{1+p/q}}{1+p/q})$ and the algebra is identical; likewise for T (with the appropriate logarithm). The *combinatorial* object χ is identical in all three cases; this is the source of equation-type independence at the constellation level.

Notation 2.6 (The classification functor). For brevity we package $F: L \mapsto \mathcal{C}(L)$ as a map from the category $\mathbf{Op}$ of scalar local irregular operators (up to formal gauge fixing the leading edge) to the category $\mathbf{MSet}_\mathbb{C}$ of finite multisets in $\mathbb{C}$ modulo global rotation. “Functor” is used here in the loose sense: the assignment respects formal gauge equivalence; functoriality with respect to morphisms in $\mathbf{Op}$ (e.g. pullback along a finite ramification, twist by an exponential) is recorded as needed below.

Symbol	Meaning
L	scalar linear operator with one irregular singularity at ∞
$\square \in \{D, S, T\}$	derivation, shift, or q -shift, respectively
$N(L)$	Newton polygon of L at ∞
$E, -p/q, r$	dominant edge; slope (lowest terms); rank multiplier
$\text{LC}_\infty(\cdot)$	leading coefficient at $x = \infty$
$\chi(c), \chi_w(w)$	WKB characteristic polynomial; inner polynomial ($\chi(c) = \chi_w(c^q)$)
$\mathcal{C}(L)$	exponent constellation = $\text{Roots}(\chi) \subset \mathbb{C}$
F	constellation map / functor $(L, \mathfrak{e}) \mapsto \mathcal{C}(L)$
μ_q, ζ_q	group of q -th roots of unity; primitive root $e^{2\pi i/q}$
Δ_{ij}	pairwise difference $c_i - c_j$, $c_i, c_j \in \mathcal{C}(L)$
$\mathcal{A}^{\text{cov}}$	anti-Stokes set on the q -fold cover $x = t^q$
$\mathbb{Z}[\zeta_q], \mathbb{Q}(\zeta_q)$	cyclotomic ring $\mathbb{Z}[\zeta_q]$ and field $\mathbb{Q}(\zeta_q)$
R_α	μ_q -ring $\{\zeta_q^j z_\alpha : 0 \leq j < q\}$

Table 1: Notation used throughout the paper.

3 The Main Theorem

We now state and prove the structural law that organises the rest of the paper. Hypotheses are kept minimal: a scalar operator with one dominant edge, in any of the three equation types.

Theorem 3.1 (Factorisation of the WKB characteristic polynomial). *Let $L = \sum_{k=0}^n a_k(x) \square^k$ be a scalar linear operator with polynomial coefficients $a_k \in \mathbb{C}[x]$ and a single irregular singularity at $x = \infty$, where $\square \in \{D, S, T\}$. Assume the Newton polygon $N(L)$ at ∞ has a unique dominant edge E of slope $-p/q$ with $\gcd(p, q) = 1$ and $p, q \geq 1$, containing $r + 1$ lattice points*

$$E \cap \mathbb{Z}^2 = \{(k_0 + jq, h_0 - jp) : 0 \leq j \leq r\}.$$

Set $\alpha_j := \text{LC}_\infty(a_{k_0+jq})$ for $0 \leq j \leq r$, and define $\chi(c)$ and $\chi_w(w)$ as in Theorem 2.4. Then

$$\boxed{\chi(c) = \chi_w(c^q),}$$

and the WKB exponent constellation factors as the μ_q -equivariant fibre product

$$\mathcal{C}(L) = \text{Roots}(\chi) = \bigsqcup_{w_i \in \text{Roots}(\chi_w)} \{c \in \mathbb{C} : c^q = w_i\},$$

where roots are counted with multiplicity and the union is over the r roots w_i of χ_w .

Proof. By Theorem 2.4, $\chi(c) = \sum_{j=0}^r \alpha_j c^{(r-j)q} = \sum_{j=0}^r \alpha_j (c^q)^{r-j} = \chi_w(c^q)$. The factorisation of $\mathcal{C}(L)$ follows by substitution: $\chi(c) = 0$ iff $c^q \in \text{Roots}(\chi_w)$, and for each such w_i the preimage under $c \mapsto c^q$ is the μ_q -orbit $\{c : c^q = w_i\}$. Multiplicities in χ_w are transmitted to χ unchanged: if w_i is a root of χ_w of multiplicity m_i , then each c with $c^q = w_i$ is a root of χ of multiplicity m_i . $\square$

Corollary 3.2 (Generic ring decomposition). *If the moduli $|w_1|, \dots, |w_r|$ are pairwise distinct and all $w_i \neq 0$, then $\mathcal{C}(L)$ consists of exactly r concentric μ_q -orbits (“rings”) of q points each, with radii $|w_i|^{1/q}$ and angular bases $\arg(w_i)/q \pmod{2\pi/q}$.*

Proof. For each simple root w_i of χ_w , the fibre $\{c : c^q = w_i\}$ consists of q points evenly spaced on the circle of radius $|w_i|^{1/q}$, with one of them at angle $\arg(w_i)/q$ and the others obtained by multiplication by ζ_q^k . The hypotheses make the r rings concentric and pairwise distinct. $\square$

Corollary 3.3 (Vieta monodromy). *Fix α_0 and let $\beta := \alpha_r$ vary. Under $\beta \mapsto e^{i\theta}\beta$,*

$$\arg\left(\prod_{c \in \mathcal{C}(L)} c\right) \mapsto \arg\left(\prod_{c \in \mathcal{C}(L)} c\right) + \theta \pmod{2\pi}.$$

If $\mathcal{C}(L)$ admits a μ_q -equivariant ring decomposition $\mathcal{C}(L) = \bigsqcup_{k=1}^s R_k$ with $|R_k| = m_k$, and the rings deform rigidly under the variation of β with rotations Δ_k (each defined modulo $2\pi/m_k$), then the unambiguous lifts $m_k \Delta_k \in \mathbb{R}/2\pi\mathbb{Z}$ obey

$$\sum_{k=1}^s m_k \Delta_k \equiv \theta \pmod{2\pi}.$$

Proof. Vieta's formula gives $\prod_{c \in \text{Roots}(\chi)} c = (-1)^{\deg \chi} \alpha_0 / \alpha_r$, so its argument is shifted by $-\theta$ under $\alpha_r \mapsto e^{i\theta} \alpha_r$ (sign convention absorbed). For the ring lift: each ring R_k contributes $m_k \Delta_k$ to the change in $\arg \prod c$ (the per-ring rotations are well-defined modulo $2\pi/m_k$, and the lift $m_k \Delta_k$ is unambiguous). Summing and matching to the total shift θ gives the rule. $\square$

Remark 3.4 (Degenerate loci). The clean ring picture of Theorem 3.2 fails on three explicit subvarieties of coefficient space $(\alpha_0, \dots, \alpha_r)$:

- (i) the *discriminant locus* $\{\text{disc} \chi_w = 0\}$, where roots of χ_w coalesce and ring multiplicities exceed q ;
- (ii) the *equal-modulus locus* $\{|w_i| = |w_j| \text{ for some } i \neq j\}$, where distinct μ_q -orbits merge into a single radius with non-uniform angular distribution;
- (iii) the *zero-root locus* $\{\alpha_r = 0\}$, where $c = 0$ enters the constellation with multiplicity q and the dominant edge itself jumps.

The first carries resonance / logarithmic Stokes data; the second is responsible for the apparent regularity of the equal-modulus stratum (e.g. a six-point constellation on one circle that is *not* a regular hexagon); the third is a wall-crossing of the Newton polygon itself. §5 treats them systematically.

Remark 3.5 (Equation-type invariance). The proof of Theorem 3.1 uses only the combinatorial identity $c^{(r-j)q} = (c^q)^{r-j}$, no analytic input. Consequently, the factorisation holds identically across D, S, T. The analytic *meaning* of $c \in \mathcal{C}(L)$ differs (it is the leading-order WKB exponent c in $f \sim \exp(cx^{p/q}/(p/q))$ for D; for S it is $\log(Sf/f)|_\infty$ to leading order; for T it is $\log_q(Tf/f)|_\infty$); the multiset $\mathcal{C}(L)$ is invariant. This is the precise sense in which F is equation-type independent.

Remark 3.6 (Forward references). Theorem 3.1 is the base case of the entire programme. It generalises in three directions: to multi-edge polygons (Theorem 4.3); to nested factorisations of χ_w (Theorem 4.6); to systems (Theorem 6.1). The complementary analytic refinement (Theorems 7.1, 7.4, 7.6 and 7.7) attaches anti-Stokes directions to the differences $\Delta_{ij} = c_i - c_j$ on the ramified cover $x = t^q$.

4 Multi-edge polygons and nested ramification

4.1 Per-edge factorisation along the slope filtration

When $N(L)$ has several edges of distinct slopes, the Main Theorem applies per edge and the constellations assemble multiplicatively via the formal-classification (Levelt–Turruttin) slope decomposition.

Lemma 4.1 (Multiplicativity of the WKB characteristic polynomial across slope-disjoint formal direct sums). *Let L_A and L_B be scalar linear operators on $\mathbb{C}[x]$ in $\{D, S, T\}$, each with a single irregular singularity at $x = \infty$. Suppose that the Newton polygon $N(L_A)$ at ∞ consists of edges $E_1^A, \dots, E_{m_A}^A$ of distinct slopes $-p_i^A/q_i^A$ ($i = 1, \dots, m_A$, $\gcd(p_i^A, q_i^A) = 1$, $p_i^A, q_i^A \geq 1$), and that $N(L_B)$ consists of a single edge of slope $-p_B/q_B$ with $\gcd(p_B, q_B) = 1$, $p_B, q_B \geq 1$. Assume the slope-disjointness hypothesis*

$$\frac{p_B}{q_B} \notin \left\{ \frac{p_1^A}{q_1^A}, \dots, \frac{p_{m_A}^A}{q_{m_A}^A} \right\}. \quad (1)$$

Let $\widehat{M}(L_A) \oplus \widehat{M}(L_B)$ denote the formal direct sum of the associated $\mathbb{C}((x^{-1}))$ -modules, and let L_{AB} be any scalar operator on $\mathbb{C}[x]$ obtained from a cyclic vector for the direct sum. Then

$$\chi_{L_{AB}}(c) = \chi_{L_A}(c) \chi_{L_B}(c),$$

where χ_{L_A}, χ_{L_B} denote the WKB characteristic polynomials of L_A, L_B (the product, in the case of L_A , of its per-edge characteristic polynomials over $E_1^A, \dots, E_{m_A}^A$).

Proof. The Newton polygon of a formal direct sum of irregular connections equals the upper convex hull of the union of the Newton polygons of the summands ([Sab07, Ch. III, §1]; or, for the classical scalar case, [Mal74, Ch. II]). Under (1), this upper hull consists of $m_A + 1$ edges, exactly one for each slope in $\{p_i^A/q_i^A\}_{i=1}^{m_A} \cup \{p_B/q_B\}$.

We analyse the LC_∞ -data along each edge E of $N(L_{AB})$. There are two cases.

Case 1: E has slope $-p_i^A/q_i^A$ for some $i \in \{1, \dots, m_A\}$. On the lattice $E \cap \mathbb{Z}^2$, only the summand $\widehat{M}(L_A)$ contributes leading-order (LC_∞) terms at slope p_i^A/q_i^A ; the summand $\widehat{M}(L_B)$ has a single slope $p_B/q_B \neq p_i^A/q_i^A$ and therefore contributes only lower-order (x^{-1} -higher-valuation) corrections to the LC_∞ -data on E . Hence

$$\text{LC}_\infty(a_k^{(AB)})|_E = \text{LC}_\infty(a_k^{(A)})|_{E_i^A} \quad \text{for } (k, h) \in E \cap \mathbb{Z}^2,$$

which by Theorem 2.4 gives the per-edge identity $\chi_{L_{AB}}^{(i,A)}(c) = \chi_{L_A}^{(i)}(c)$.

Case 2: E has slope $-p_B/q_B$. By the same argument with the roles of L_A and L_B exchanged, the LC_∞ -data on E is contributed solely by the (single edge of) $\widehat{M}(L_B)$. Hence $\chi_{L_{AB}}^{(B)}(c) = \chi_{L_B}(c)$.

Multiplying over all edges of $N(L_{AB})$ and using that for a scalar operator with multiple slope-distinct edges the total WKB characteristic polynomial is the product of its per-edge characteristic polynomials,

$$\chi_{L_{AB}}(c) = \left(\prod_{i=1}^{m_A} \chi_{L_{AB}}^{(i,A)}(c) \right) \chi_{L_{AB}}^{(B)}(c) = \left(\prod_{i=1}^{m_A} \chi_{L_A}^{(i)}(c) \right) \chi_{L_B}(c) = \chi_{L_A}(c) \chi_{L_B}(c),$$

as claimed. $\square$

Remark 4.2 (Iteration over arbitrarily many slope-disjoint summands). The single-slope restriction on L_B is essential only in the inductive step; *either* operand may be multi-slope provided the slope-disjointness hypothesis (1) (interpreted as disjointness of slope sets) holds. In particular, an obvious induction on the number of slopes establishes the multi-fold multiplicativity statement

$$\chi_{L_1 \oplus \dots \oplus L_m}(c) = \prod_{\alpha=1}^m \chi_{L_\alpha}(c),$$

where each L_α is single-slope of slope $-p_\alpha/q_\alpha$ and the slopes are pairwise distinct. This is exactly the form invoked in the proof of Theorem 4.3 (Step 3): one applies the lemma to $(\widehat{M}_1 \oplus \dots \oplus \widehat{M}_{m-1}) \oplus \widehat{M}_m$, with $L_A := L_1 \oplus \dots \oplus L_{m-1}$ a multi-slope operator (with slope set $\{p_1/q_1, \dots, p_{m-1}/q_{m-1}\}$) and $L_B := L_m$ single-slope with slope p_m/q_m distinct from those of L_A . The slope-disjointness hypothesis is exactly the multi-slope hypothesis $p_i/q_i \neq p_j/q_j$ for $i \neq j$ in the parent operator L .

Theorem 4.3 (Per-edge factorisation; multi-edge generalisation). *Let L be a scalar linear operator with polynomial coefficients on $\mathbb{C}[x]$ and a single irregular singularity at $x = \infty$. Suppose $N(L)$ has edges $E_1, \dots, E_m$ with pairwise distinct slopes $-p_\alpha/q_\alpha$ ($\gcd(p_\alpha, q_\alpha) = 1$) and $r_\alpha + 1$ lattice points on E_α . Let $k_0^{(\alpha)}$ be the height of E_α above the c -axis, and let $\chi_w^{(\alpha)} \in \mathbb{C}[w]$ be the inner polynomial of degree r_α built from the LC_∞ data of E_α (per Theorem 2.4). Then:*

1. (Slope decomposition.) *The formal differential module $\widehat{M}(L)$ associated to L at $x = \infty$ admits a canonical direct-sum decomposition $\widehat{M}(L) \simeq \bigoplus_{\alpha=1}^m \widehat{M}_\alpha$ with $\widehat{M}_\alpha$ of pure slope p_α/q_α . Choosing cyclic vectors realises each $\widehat{M}_\alpha$ as the formal module of a scalar single-slope operator L_α , and we write $\widehat{L} \simeq L_1 \oplus \dots \oplus L_m$.*
2. (Per-edge characteristic polynomial.) *Each L_α satisfies Theorem 3.1 with $\chi^{(\alpha)}(c) = c^{k_0^{(\alpha)}} \chi_w^{(\alpha)}(c^{q_\alpha})$.*
3. (Total factorisation.) *Setting $\chi(c) := \prod_{\alpha=1}^m \chi^{(\alpha)}(c)$, one has $\mathcal{C}(L) = \bigsqcup_{\alpha=1}^m \mathcal{C}_\alpha$ as multisets, with $\mathcal{C}_\alpha := \text{Roots}(\chi^{(\alpha)})$.*

On the open subset U_{disj} of coefficient space where the per-edge radial annuli $[\rho_\alpha^{\min}, \rho_\alpha^{\max}]$ with $\rho_\alpha^\bullet = |w^\bullet|^{1/q_\alpha}$ are pairwise disjoint, the union is a disjoint union of distinct subsets of $\mathbb{C}$.

Proof of Theorem 4.3 (Per-edge factorisation; multi-edge generalisation). We organise the argument in four steps.

Step 1 (Slope decomposition; Levelt–Turrittin). By the Levelt–Turrittin theorem (Sibuya [Sib90]; Wasow [Was87, Ch. VII]; Sabbah [Sab07, Ch. II, Thm. II.2.4]), the formal $\mathbb{C}((x^{-1}))$ -module $\widehat{M}(L)$ associated to L at $x = \infty$ admits a canonical slope decomposition

$$\widehat{M}(L) = \bigoplus_{\alpha=1}^m \widehat{M}_\alpha,$$

where $\widehat{M}_\alpha$ is pure of slope p_α/q_α and the multiset $\{p_\alpha/q_\alpha\}_{\alpha=1}^m$ is exactly the set of slopes of the edges of $N(L)$. The decomposition is canonical (unique up to gauge), and choosing a cyclic vector in each summand $\widehat{M}_\alpha$ realises it as the formal module of a scalar single-slope operator L_α . The Newton polygon of L_α consists of a single edge of slope $-p_\alpha/q_\alpha$ with $r_\alpha + 1$ lattice points, matching the corresponding edge E_α of $N(L)$.

Step 2 (Per-edge Main Theorem). Each L_α has Newton polygon $N(L_\alpha)$ a single edge of slope $-p_\alpha/q_\alpha$ with $r_\alpha + 1$ lattice points; Theorem 3.1 applies and gives the explicit factorisation

$$\chi_{L_\alpha}(c) = c^{k_0^{(\alpha)}} \chi_w^{(\alpha)}(c^{q_\alpha}),$$

where $k_0^{(\alpha)}$ is the multiplicity of $c = 0$ in χ_{L_α} contributed by lattice points strictly below E_α (the “foot” of the edge in the scalar operator L), and $\chi_w^{(\alpha)} \in \mathbb{C}[w]$ is the inner polynomial of degree r_α built from the LC_∞ -data of E_α per Theorem 2.4.

Step 3 (Multiplicativity across the slope filtration). Apply Theorem 4.1 inductively to the m -fold formal direct sum

$$\widehat{M}(L) = \widehat{M}_1 \oplus \dots \oplus \widehat{M}_m$$

(equivalently, $\widehat{L} \simeq L_1 \oplus \dots \oplus L_m$, in the sense of Step 1). The $m = 2$ case is Theorem 4.1 directly; the inductive step is the identity $\widehat{M}_1 \oplus \dots \oplus \widehat{M}_m = (\widehat{M}_1 \oplus \dots \oplus \widehat{M}_{m-1}) \oplus \widehat{M}_m$ together with the single-slope hypothesis on $\widehat{M}_m$ (its single slope p_m/q_m is distinct from those of the first $m - 1$ summands by hypothesis on L). Hence

$$\chi_L(c) = \prod_{\alpha=1}^m \chi_{L_\alpha}(c) = \prod_{\alpha=1}^m c^{k_0^{(\alpha)}} \chi_w^{(\alpha)}(c^{q_\alpha}).$$

Taking roots with multiplicity,

$$\mathcal{C}(L) = \text{Roots}(\chi_L) = \bigsqcup_{\alpha=1}^m \text{Roots}(\chi_{L_\alpha}) = \bigsqcup_{\alpha=1}^m \mathcal{C}_\alpha$$

as multisets.

Step 4 (Genericity). On the open subset U_{disj} of coefficient space where the per-edge radial annuli

$$[\rho_\alpha^{\min}, \rho_\alpha^{\max}], \quad \rho_\alpha^\bullet = \min_w / \max_w \{|w|^{1/q_\alpha} : w \in \text{Roots}(\chi_w^{(\alpha)})\},$$

are pairwise disjoint as subsets of $\mathbb{R}_{>0}$, the multiset union $\bigsqcup_\alpha \mathcal{C}_\alpha$ is also an honest disjoint union of subsets of $\mathbb{C}^\times$, because the per-edge constellations occupy disjoint moduli. On the complement (the radii-collision stratum), the multiset disjoint union is preserved by Step 3; only the set-theoretic statement in $\mathbb{C}^\times$ is weakened, namely, distinct edges may contribute roots of equal absolute value. $\square$

Remark 4.4. The choice of cyclic vector in Step 1 affects the scalar operator L_α only up to a formal gauge equivalence; this gauge preserves χ_{L_α} (Theorem 1.2), so the per-edge characteristic polynomials $\chi^{(\alpha)}$ depend canonically on $\bar{M}_\alpha$ and hence on the edge E_α . The statement of Theorem 4.3 is therefore intrinsic.

Remark 4.5 (Counterexample boundary for M1). The disjoint-union conclusion in (3) is a multiset statement; as *sets* in $\mathbb{C}^\times$ the per-edge constellations may coincide on the *inter-edge equal-radius stratum*, where the radial annuli of two distinct edges accidentally share a radius. Examples are documented in the companion dataset [Pap26]. Likewise, when a per-edge $\chi_w^{(\alpha)}(0) = 0$, the zero-root contributions of adjacent edges *share* the origin in $\mathbb{C}$. These are not contradictions: the multiset disjoint union is preserved; only the set-theoretic statement is weakened.

4.2 Nested ramification

Beyond a single μ_q -pullback, the inner polynomial χ_w itself may factor through a higher power.

Theorem 4.6 (Nested ramification: iterated μ_q -pullback). *Let L be scalar with one dominant edge of slope $-p/q$ as in Theorem 3.1. Suppose the inner polynomial admits a factorisation $\chi_w(w) = \chi_v(w^{q'})$ for some integer $q' \geq 2$ and a polynomial $\chi_v \in \mathbb{C}[v]$ of degree $r' = r/q'$ (i.e. χ_w has nonzero coefficients only in degrees that are multiples of q'). Then*

$$\mathcal{C}(L) = \bigsqcup_{v \in \text{Roots}(\chi_v)} \{c \in \mathbb{C} : c^{qq'} = v\}.$$

Each v -fibre $\{c : c^{qq'} = v\}$ has size qq' and carries a free $\mu_{qq'}$ -action. The visible symmetry group generated by the outer μ_q -action (from $\chi(c) = \chi_w(c^q)$) and the nested $\mu_{q'}$ -action (from $\chi_w = \chi_v \circ (\cdot)^{q'}$) is

$$\mu_{\text{lcm}(q,q')} \subseteq \mu_{qq'},$$

with equality $\mu_{\text{lcm}(q,q')} = \mu_{qq'}$ if and only if $\gcd(q,q') = 1$ (in which case $\mu_{qq'} \cong \mu_q \times \mu_{q'}$ canonically by CRT).

Proof. By Theorem 3.1, $\chi(c) = \chi_w(c^q)$; substituting the hypothesis gives $\chi(c) = \chi_v((c^q)^{q'}) = \chi_v(c^{qq'})$. The fibre decomposition follows. For the symmetry claim, the original Main Theorem's μ_q -action sits inside $\mu_{qq'}$ as the subgroup $\{\zeta_{qq'}^{q'k}\}$. A lift to c -level of the inner-polynomial $\mu_{q'}$ -symmetry (acting on $w = c^q$ by $w \mapsto \zeta_{q'} w$) is multiplication by an $\eta \in \mu_{qq'}$ with $\eta^q = \zeta_{q'}$; such η exists (e.g. $\eta = \zeta_{qq'}^q$) and is unique modulo μ_q . The subgroup of $\mu_{qq'}$ generated by ζ_q and η is $\mu_{\text{lcm}(q,q')}$. $\square$

Remark 4.7 (Minimality). The hypothesis $\chi_w(w) = \chi_v(w^{q'})$ is preserved under further inner factorisations $\chi_v(v) = \chi_u(v^{q''})$; iterating until no inner factorisation is possible gives the *minimal* $q'q'' \cdots$ through which the constellation factors. The companion atlas [Pap26] records two such nested operators, both stable under one round of inner factorisation.

Remark 4.8 (Constellation vs. analytic ramification). Theorem 4.6 is a purely algebraic identity at the constellation level. The analytic implications for the Levelt–Turrittin ramification cover are deferred to Theorem 7.4; crucially, the minimal cover for the formal classification remains $x = t^q$ and does not increase to $x = \tau^{qq'}$. The nested factorisation refines the wild character variety, not the ramification index.

5 Degeneracy strata and arithmetic stratification

The Main Theorem (Theorem 3.1) decomposes the image of F into a generic stratum and the three degenerate loci of Theorem 3.4. We now treat the degeneracy strata and their arithmetic refinement, culminating in a prime- q Kummer dichotomy (Theorem 5.1).

5.1 The four named strata

In the companion atlas [Pap26], four stress-test operators exhibit the qualitatively distinct degenerate behaviours:

1. *Equal-modulus locus*. Six points on a single circle of radius $\sqrt{2}$, *not* evenly spaced.
2. *Discriminant locus* (resonant multiplicity): two rings, the inner with multiplicity 2 at each apparent point.
3. *Zero-root locus* (wall-crossing): two points at radius 1 together with a multiplicity- q root at the origin.
4. *Accidental-symmetry locus*. A regular rq -gon with cyclic symmetry $C_{rq} \supsetneq C_q$, occurring at slope $-p/q$ but exhibiting symmetry above what q predicts.

The Vieta lift (Theorem 3.3) characterises the equal-modulus stratum behaviour (rigid permutation up to shared rotation, not rigid rotation). The discriminant locus carries logarithmic / resonant Stokes data (Theorem 3.4(i)). The zero-root locus is a wall-crossing of $N(L)$ itself: when $\alpha_r = 0$ the dominant edge has been mis-identified and is replaced by a strictly shorter edge with smaller r .

5.2 The Kummer dichotomy for prime q

Beyond the open stratum, the arithmetic of the inner polynomial controls whether the per-ring constellations admit nontrivial integer linear relations. The classical setting is Kummer theory: when ratios of inner roots are q -th powers in $\mathbb{Q}(\zeta_q)$, the constellation collapses arithmetically.

Theorem 5.1 (Inner-polynomial Kummer dichotomy; prime q , revised). *Let L be a scalar linear operator on $\mathbb{C}[x]$ with a single irregular singularity at $x = \infty$ and a single dominant Newton edge of slope $-p/q$ ($\gcd(p, q) = 1$), with q prime. Suppose the inner polynomial $\chi_w \in \mathbb{C}[w]$ is squarefree with $r \geq 2$ distinct nonzero roots*

$$\text{Roots}(\chi_w) = \{w_1, \dots, w_r\} \subset \mathbb{C}^\times.$$

Fix a consistent choice of q -th roots $z_\alpha \in \mathbb{C}$ with $z_\alpha^q = w_\alpha$, so that

$$\mathcal{C}(L) = \bigsqcup_{\alpha=1}^r R_\alpha, \quad R_\alpha = \{ \zeta_q^j z_\alpha : 0 \leq j < q \}.$$

For $\mathbb{Z}$ -coefficients $n_{\alpha j} \in \mathbb{Z}$, set

$$\lambda_{\alpha} := \sum_{j=0}^{q-1} n_{\alpha j} \zeta_q^j \in \mathbb{Z}[\zeta_q].$$

Call a $\mathbb{Z}$ -relation $\sum_{\alpha,j} n_{\alpha j} \zeta_q^j z_{\alpha} = 0$ μ_q -forced if $\lambda_{\alpha} = 0$ in $\mathbb{Z}[\zeta_q]$ for every α , and cross-ring nontrivial otherwise. Then:

1. **Algebraic-independence stratum.** If $\{w_1, \dots, w_r\}$ are algebraically independent over $\mathbb{Q}$, then $\mathcal{C}(L)$ admits no cross-ring nontrivial $\mathbb{Z}$ -relation (no bound on the support or coefficient size of the relation is required).

2. **Kummer stratum.** If there exist distinct $\alpha, \beta \in \{1, \dots, r\}$ and $u \in \mathbb{Q}(\zeta_q)^{\times}$ with

$$w_{\beta} = u^q w_{\alpha},$$

then, after suitable consistent choices of q -th roots, the relation $z_{\beta} = u z_{\alpha}$ holds and is a cross-ring nontrivial $\mathbb{Z}[\zeta_q]$ -relation; when $u \in \mathbb{Q}^{\times}$, it gives a cross-ring nontrivial $\mathbb{Z}$ -relation of support 2 on $\{z_{\alpha}, z_{\beta}\}$.

Statements (1) and (2) are mutually exclusive: an operator in the algebraic-independence stratum has no Kummer relation of the form (2), and conversely a Kummer relation contradicts algebraic independence over $\mathbb{Q}$.

Proof of Theorem 5.1. Set-up. Rearranging $\sum_{\alpha,j} n_{\alpha j} \zeta_q^j z_{\alpha} = 0$ by ring index gives

$$\sum_{\alpha=1}^r \lambda_{\alpha} z_{\alpha} = 0 \quad \text{in } \mathbb{C}, \quad \lambda_{\alpha} \in \mathbb{Z}[\zeta_q]. \quad (2)$$

Kernel of $\mathbb{Z}^q \twoheadrightarrow \mathbb{Z}[\zeta_q]$. For prime q , the cyclotomic ring $\mathbb{Z}[\zeta_q] = \mathbb{Z}[x]/\Phi_q(x)$ is a free $\mathbb{Z}$ -module of rank $q-1$, with $\mathbb{Z}$ -basis $\{1, \zeta_q, \dots, \zeta_q^{q-2}\}$. The surjection

$$\pi: \mathbb{Z}^q \twoheadrightarrow \mathbb{Z}[\zeta_q], \quad (n_0, \dots, n_{q-1}) \mapsto \sum_j n_j \zeta_q^j,$$

has kernel the rank-one $\mathbb{Z}$ -submodule generated by the cyclotomic relation

$$(1, 1, \dots, 1) \in \mathbb{Z}^q, \quad \pi(1, \dots, 1) = 1 + \zeta_q + \dots + \zeta_q^{q-1} = 0.$$

Hence $\lambda_{\alpha} = 0$ in $\mathbb{Z}[\zeta_q]$ if and only if the vector $(n_{\alpha 0}, \dots, n_{\alpha, q-1})$ is a constant vector $(c_{\alpha}, \dots, c_{\alpha})$ for some $c_{\alpha} \in \mathbb{Z}$. In this case

$$\sum_j n_{\alpha j} \zeta_q^j z_{\alpha} = c_{\alpha} \left(\sum_j \zeta_q^j \right) z_{\alpha} = c_{\alpha} \cdot 0 = 0,$$

giving the per-ring trivial relations. Thus μ_q -forced relations are exactly the integer combinations of the r per-ring sums $\sum_{c \in R_{\alpha}} c = 0$.

Step 1 (Algebraic independence implies μ_q -forced). Assume $\{w_1, \dots, w_r\}$ are algebraically independent over $\mathbb{Q}$; then they are algebraically independent over $\mathbb{Q}(\zeta_q)$ as well (an algebraic relation over $\mathbb{Q}(\zeta_q)$ would specialise, on choosing an embedding $\mathbb{Q}(\zeta_q) \hookrightarrow \overline{\mathbb{Q}}$, to an algebraic relation over $\mathbb{Q}$ in finitely many w_{α} , contradicting algebraic independence).

We show by induction on r that the field tower

$$\mathbb{Q}(\zeta_q) \subset \mathbb{Q}(\zeta_q)(w_1, \dots, w_r) \subset \mathbb{Q}(\zeta_q)(w_1, \dots, w_r)(z_1) \subset \dots \subset \mathbb{Q}(\zeta_q)(w_1, \dots, w_r)(z_1, \dots, z_r) \quad (3)$$

has degrees

$$[\mathbb{Q}(\zeta_q)(w_1, \dots, w_r)(z_1, \dots, z_\alpha) : \mathbb{Q}(\zeta_q)(w_1, \dots, w_r)(z_1, \dots, z_{\alpha-1})] = q$$

for each $\alpha = 1, \dots, r$.

The polynomial $X^q - w_\alpha$ has degree q over its coefficient field. By Kummer theory ([Lan02, Ch. VI, Thm. 9.1]), the polynomial $X^q - w_\alpha$ is irreducible over a field $K \supseteq \mathbb{Q}(\zeta_q)$ if and only if w_α is not a q -th power in K . We apply this with $K = \mathbb{Q}(\zeta_q)(w_1, \dots, w_r)(z_1, \dots, z_{\alpha-1})$. By the inductive hypothesis $[K : \mathbb{Q}(\zeta_q)(w_1, \dots, w_r)] = q^{\alpha-1}$, and the algebraic independence of the w_β over $\mathbb{Q}(\zeta_q)$ guarantees that w_α is transcendental over $\mathbb{Q}(\zeta_q)(w_1, \dots, \widehat{w_\alpha}, \dots, w_r)$. In particular, w_α is not a q -th power in K (a q -th power in K would be algebraic over $\mathbb{Q}(\zeta_q)(w_1, \dots, \widehat{w_\alpha}, \dots, w_r, z_1, \dots, z_{\alpha-1})$, but this latter field is algebraic over $\mathbb{Q}(\zeta_q)(w_1, \dots, \widehat{w_\alpha}, \dots, w_r)$ since each z_β is). Hence $X^q - w_\alpha$ is irreducible over K , the degree is q , and the inductive step is complete.

It follows that $[\mathbb{Q}(\zeta_q)(w_1, \dots, w_r)(z_1, \dots, z_r) : \mathbb{Q}(\zeta_q)(w_1, \dots, w_r)] = q^r$, and the set $\{z_1, \dots, z_r\}$ is algebraically independent over $\mathbb{Q}(\zeta_q)(w_1, \dots, w_r)$ (an algebraic relation would lower the transcendence degree). In particular, the z_α are linearly independent over $\mathbb{Q}(\zeta_q)$ (hence over $\mathbb{Z}[\zeta_q]$).

Suppose now that (2) holds with not all $\lambda_\alpha = 0$ in $\mathbb{Z}[\zeta_q]$. Then $\sum_\alpha \lambda_\alpha z_\alpha = 0$ is a $\mathbb{Q}(\zeta_q)$ -linear dependence among the z_α , contradicting the previous paragraph. Hence all $\lambda_\alpha = 0$ in $\mathbb{Z}[\zeta_q]$, and the $\mathbb{Z}$ -relation is μ_q -forced. This proves (1).

Step 2 (Kummer relation gives a cross-ring relation). Suppose $w_\beta = u^q w_\alpha$ for some $u \in \mathbb{Q}(\zeta_q)^\times$ and distinct $\alpha \neq \beta$. Choose $z_\alpha \in \mathbb{C}$ with $z_\alpha^q = w_\alpha$ and set $z_\beta := u z_\alpha$; then $z_\beta^q = u^q w_\alpha = w_\beta$ as required. The identity $z_\beta - u z_\alpha = 0$ is a $\mathbb{Z}[\zeta_q]$ -relation with $\lambda_\alpha = -u \neq 0$ in $\mathbb{Z}[\zeta_q]$ and $\lambda_\beta = 1 \neq 0$ in $\mathbb{Z}[\zeta_q]$, so it is cross-ring nontrivial.

If in addition $u \in \mathbb{Q}^\times$, write $u = a/b$ in lowest terms with $a \in \mathbb{Z}$, $b \in \mathbb{Z}_{>0}$, and $\gcd(a, b) = 1$. Then clearing denominators in $z_\beta - u z_\alpha = 0$ gives the genuine $\mathbb{Z}$ -relation

$$b z_\beta - a z_\alpha = 0 \quad \text{in } \mathbb{C}, \quad (4)$$

which has support 2 on $\{z_\alpha, z_\beta\}$ and integer coefficients of absolute value $\max(|a|, b)$. Explicitly, the $\mathbb{Z}$ -coefficient pattern from (2) is

$$n_{\alpha 0} = -a, \quad n_{\beta 0} = b, \quad n_{\gamma j} = 0 \quad \text{for all other } (\gamma, j),$$

so $\lambda_\alpha = -a$, $\lambda_\beta = b$, neither of which is zero in $\mathbb{Z}[\zeta_q]$. This establishes a cross-ring nontrivial $\mathbb{Z}$ -relation of support two and bounded coefficient size that is detectable by integer-relation searches with window $\geq \max(|a|, b)$. This proves (2).

Step 3 (Mutual exclusivity). A Kummer relation $w_\beta = u^q w_\alpha$ with $u \in \mathbb{Q}(\zeta_q)^\times$ exhibits w_β as an algebraic function of w_α over $\mathbb{Q}(\zeta_q)$ (in fact a polynomial relation $w_\beta = u^q w_\alpha$ in $\mathbb{Q}(\zeta_q)[w_\alpha]$), hence over $\mathbb{Q}$ (since $\mathbb{Q}(\zeta_q) \subset \overline{\mathbb{Q}}$). This is incompatible with the algebraic independence of $\{w_1, \dots, w_r\}$ over $\mathbb{Q}$. Hence (1) and (2) cannot hold simultaneously.

Scope: the dichotomy is not a trichotomy. The two strata are mutually exclusive but *not* exhaustive. There exist single-edge prime- q operators whose inner-polynomial roots $\{w_1, \dots, w_r\}$ admit a nontrivial algebraic relation over $\mathbb{Q}$ that is not of the pure Kummer form $w_\beta = u^q w_\alpha$ (for example, multiplicative dependencies $w_1^a w_2^b = w_3^c$ with $(a, b, c) \neq (0, 0, q)$ modulo per-variable Kummer reductions, or transcendence-defect relations among three or more w_α). On such operators Theorem 5.1 is silent: neither case (1) nor case (2) applies, and the cross-ring relation structure of $\mathcal{C}(L)$ is governed by the specific shape of the algebraic dependence rather than by Kummer theory. This stratum is the “residual” between the open algebraic-independence locus and the Kummer locus; it is closed in the space of inner polynomials but strictly smaller than the algebraic-dependence locus. $\square$

Remark 5.2 (Composite q). For composite q , $\mathbb{Z}[\zeta_q]$ has further structure: the kernel of the projection $\mathbb{Z}[\zeta_q] \rightarrow \prod_{d|q} \mathbb{Z}[\zeta_d]$ captures sub-cyclotomic resonances, and the μ_q -forced sub-lattice

includes sub-orbit sums coming from each divisor $d \mid q$. The corresponding dichotomy holds with “ $\lambda_\alpha = 0$ in $\mathbb{Z}[\zeta_q]$ ” replaced by “ λ_α in the kernel of the projection”. Examples implicit in $q = 4$ and $q = 6$ operators of the atlas illustrate the refinement.

Remark 5.3 (Bounded-search corroboration). The companion dataset [Pap26] contains a sweep of all single-edge prime- q operators with $q \in \{5, 7\}$ in the single-edge (23-class) atlas. For every such operator, an integer relation search with $|n_k| \leq 3$ and support ≤ 4 produces zero cross-ring relations on $\mathcal{C}(L)$ – consistent with the algebraic statement Theorem 5.1(1). For every multi-ring operator placed in the Kummer stratum (rational ratios of the w_α), the search detects at least five cross-ring relations within the same window – consistent with Theorem 5.1(2). The search-window data therefore *corroborate* the algebraic dichotomy on the atlas; the algebraic statement itself does not depend on the search window.

6 Systems

The constellation functor extends to systems whose leading symbol is semisimple, via a block-diagonal pullback.

Theorem 6.1 (System pullback: block-diagonal direct sum). *Let $\Phi = \text{diag}(L_1, \dots, L_n)$ be a block-diagonal system of scalar linear operators, each L_i scalar with a single irregular singularity at $x = \infty$ and Newton polygon $N(L_i)$. Then*

$$\mathcal{C}(\Phi) = \bigsqcup_{i=1}^n \mathcal{C}(L_i) \quad \text{as multisets,}$$

and the constellation functor factors

$$\mathbf{F}: \text{Sys}^n(\text{scalar}) \longrightarrow \text{Sym}^n(\text{finite multisets in } \mathbb{C}), \quad \mathbf{F}(\Phi) = \bigsqcup_i \mathbf{F}(L_i).$$

Proof. For a block-diagonal system, the WKB symbol of Φ acts diagonally on each block, so the leading balance decouples and the WKB characteristic polynomial of Φ is $\chi_\Phi(c) = \prod_i \chi_{L_i}(c)$. Hence $\text{Roots}(\chi_\Phi) = \bigsqcup_i \text{Roots}(\chi_{L_i})$ as multisets. The factorisation through the symmetric power is the observation that the constellation is a multiset, not an ordered tuple, so the assignment $\Phi \mapsto (\mathcal{C}(L_1), \dots, \mathcal{C}(L_n))$ is $\mathfrak{S}_n$ -equivariant. $\square$

Remark 6.2 (Diagonalisable coupled systems). For a general first-order system whose leading symbol $A_0(x) = \text{LC}_\infty(\Phi)$ is diagonalisable with distinct eigenvalues, a formal gauge reduces Φ to block-diagonal form to leading order, and Theorem 6.1 applies at the constellation level. Coalescence of leading eigenvalues defines a *system discriminant stratum* where the Levelt–Turrittin classification of irregular formal types is required; the present theorem does not extend to that stratum without further hypotheses, and genuinely coupled systems (non-semisimple leading symbol, Jordan blocks) lie outside the present scope. See Boalch [Boa14, Boa16] for the moduli-theoretic treatment.

Remark 6.3 (Counterexample boundary for S1). The companion counterexample catalogue (CE-coupled-system) records that generic Painlevé-type and coupled-system examples lie outside the block-diagonal hypothesis; in such cases the leading-symbol diagonalisation is itself a non-trivial gauge problem whose obstruction sits in the system-discriminant stratum.

7 Analytic lifting: anti-Stokes skeleton and slope filtration

7.1 Open-stratum hypotheses

The analytic theorems below organise the Stokes geometry attached to a constellation. They are clearest on the *open stratum* where no degeneracies of Theorem 3.4 occur. We collect the running hypotheses:

(OS1) χ_w squarefree ($\text{disc}\chi_w \neq 0$).

(OS2) $\chi_w(0) \neq 0$ (no zero roots in $\mathcal{C}(L)$).

(OS3) *Sharpened*. No ω -resonance beyond μ_q : the multiset $\{\arg \Delta_{ij} \bmod \pi/p\}$ has no coincidences beyond those forced by the μ_q -ring structure on $\mathcal{C}(L)$. Equivalently, every coincidence $\arg \Delta_{ij} \equiv \arg \Delta_{i'j'} \pmod{\pi/p}$ must arise from a pairing of unordered μ_q -images $\{c_i, c_j\} \subset R_\alpha$ and $\{c_{i'}, c_{j'}\} \subset R_\alpha$ inside the SAME ring R_α ; cross-ring coincidences are excluded.

(OS4) (Multi-edge.) Per-edge radial annuli pairwise disjoint.

(OS5) (Multi-edge.) Cross-edge leading coefficients nonzero ($\chi_w^{(\beta)}(0) \neq 0$ for each β).

Theorem 7.1 uses (OS1)–(OS3); Theorem 7.4 uses (OS1)–(OS3); Theorem 7.6 uses (OS1)–(OS5); Theorem 7.7 uses (OS1)–(OS3).

7.2 Anti-Stokes directions and $\arg \Delta_{ij}$

Theorem 7.1 (Anti-Stokes $\leftrightarrow \arg \Delta_{ij}$ on the cover; open stratum). *Let L be scalar with one dominant edge of slope $-p/q$, $\square = D$, in the open stratum (OS1)–(OS3). Let $\mathcal{C}(L) = \{c_1, \dots, c_N\}$ with $N = qr$ and $\Delta_{ij} = c_i - c_j$. On the ramified cover $x = t^q$, the formal connection splits as a direct sum*

$$\widehat{L}|_{x=t^q} \simeq \bigoplus_{i=1}^N (\mathbb{C}\{t\}, d + dQ_i(t)), \quad Q_i(t) = c_i t^p + R_i(t), \quad \deg R_i < p,$$

and the anti-Stokes directions on the t -plane are

$$\mathcal{A}^{\text{cov}} = \{ \phi \in [0, 2\pi) : \arg(\Delta_{ij}) + p\phi \equiv \frac{\pi}{2} \pmod{\pi}, \text{ some unordered } \{i, j\} \}.$$

Each unordered pair $\{i, j\}$ contributes exactly $2p$ cover-rays; the total cover-ray count with multiplicity is $2p\binom{N}{2}$. Under the projection $t \mapsto t^q$, each pair's $2p$ cover-rays project to $2p/\gcd(2p, q)$ base-rays.

Proof of Theorem 7.1. We work throughout under (OS1)–(OS3), with $\square = D$, the open stratum hypotheses collected in Section 7.1.

Step 1 (Splitting on the q -cover by Levelt–Turrittin). The slope p/q of the dominant edge has denominator q ; by the Levelt–Turrittin theorem (Sibuya [Sib90]; Wasow [Was87, Ch. VII, Thms. 19.1, 19.2]; Sabbah [Sab07, Ch. II, Thm. II.2.4]), the formal connection $\widehat{L}$ at $x = \infty$ becomes gauge-equivalent, on a ramification cover of degree q (here, $x = t^q$), to a direct sum of formal rank-one summands:

$$\widehat{L}|_{x=t^q} \simeq_{\text{formal}} \bigoplus_{i=1}^N (\mathbb{C}((t^{-1})), d + dQ_i(t)), \quad N = qr,$$

with $Q_i(t) \in t^{-1}\mathbb{C}[t]$ of degree at most p in t , and (crucially) leading term

$$Q_i(t) = c_i t^p + R_i(t), \quad R_i \in \mathbb{C}[t], \quad \deg R_i < p,$$

where the leading coefficient c_i is precisely the i -th element of $\mathcal{C}(L) = \text{Roots}(\chi)$. That the leading coefficients are roots of χ follows from the WKB leading-balance characterisation of the irregular type ([Sab07, Ch. II, §1.3]); under (OS1)–(OS2) (squarefreeness of χ_w and absence of zero roots), the c_i are pairwise distinct and nonzero.

Step 2 (Anti-Stokes condition: leading-term form). For the rank-one summand $(\mathbb{C}((t^{-1})), d + dQ_i)$, a horizontal section has formal expression $f_i(t) \sim \exp(-Q_i(t))$ times a formal Laurent series of moderate growth. Comparing two summands $i \neq j$ along a ray $\arg t = \phi$, the dominance relation $|f_i(t)/f_j(t)|$ is controlled by $\operatorname{Re}(Q_j(t) - Q_i(t))$ at large $|t|$. The (anti-)Stokes directions for the pair (i, j) are the rays ϕ along which the leading term of $\operatorname{Re}(Q_i - Q_j)$ vanishes in t , that is, where dominance flips between the two summands.

For pair (i, j) , the difference of irregular types is

$$Q_i(t) - Q_j(t) = (c_i - c_j)t^p + (R_i - R_j), \quad \deg(R_i - R_j) < p.$$

On the open stratum, (OS1) gives $c_i \neq c_j$ (squarefreeness of χ_w separates the roots of χ), so $\Delta_{ij} = c_i - c_j \in \mathbb{C}^\times$ and the leading t^p -term in $Q_i - Q_j$ is nonzero.

Step 3 (Real-part calculation). Write $t = \rho e^{i\phi}$ with $\rho > 0$ and $\phi \in [0, 2\pi)$, and $\Delta_{ij} = |\Delta_{ij}| e^{i \arg \Delta_{ij}}$. Then

$$\Delta_{ij} t^p = |\Delta_{ij}| \rho^p e^{i(\arg \Delta_{ij} + p\phi)},$$

and

$$\operatorname{Re}(Q_i - Q_j)(\rho e^{i\phi}) = |\Delta_{ij}| \rho^p \cos(\arg \Delta_{ij} + p\phi) + O(\rho^{p-1}).$$

The leading ρ^p -term vanishes precisely when

$$\arg \Delta_{ij} + p\phi \equiv \frac{\pi}{2} \pmod{\pi},$$

which is the anti-Stokes condition stated in the theorem. The solutions in $[0, 2\pi)$ form an arithmetic progression with common difference π/p , hence exactly $2p$ rays per unordered pair $\{i, j\}$.

Step 4 (Total count and distinctness). Summing the per-pair counts over all $\binom{N}{2}$ unordered pairs gives a total of $2p\binom{N}{2}$ cover-rays *with multiplicity*. The number of *distinct* cover-rays is governed by collisions in the multiset

$$\{ \arg \Delta_{ij} \pmod{\pi/p} : \{i, j\} \subset \{1, \dots, N\} \}.$$

Under (OS3), no such collisions occur beyond those forced by the ring structure, so the number of distinct cover-rays is the full $2p\binom{N}{2}$ count modulo the ring-forced identifications.

Step 5 (Projection cover-to-base). The cover-to-base map $t \mapsto t^q$ acts on phases by $\phi \mapsto q\phi \pmod{2\pi}$. The image of a fixed pair's $2p$ cover-rays (an arithmetic progression of spacing π/p) under multiplication by q is an arithmetic progression of spacing $q\pi/p$. Reducing modulo 2π , this progression has period $\gcd(2p, q)\pi/p$, and the number of distinct points modulo 2π is $2p/\gcd(2p, q)$, with each base-ray covered by $\gcd(2p, q)$ cover-rays. This is the projection formula stated in the theorem.

Step 6 (Determinacy: Δ_{ij} -geometry determines rays). The set of anti-Stokes rays for the pair (i, j) is the arithmetic progression

$$\phi_k(i, j) = \frac{1}{p} \left(\frac{\pi}{2} - \arg \Delta_{ij} \right) + \frac{k\pi}{p}, \quad k = 0, 1, \dots, 2p-1,$$

depending only on $\arg \Delta_{ij} \pmod{\pi/p}$ and p . The multiset of all such rays for all $\binom{N}{2}$ pairs is therefore recovered from the multiset $\{(\arg \Delta_{ij}, p) : i < j\}$. This proves the final claim: $\mathcal{A}^{\text{cov}}$ is determined as a multiset (with multiplicity-by-pair-count) by the Δ_{ij} -geometry of the constellation on the cover. $\square$

Remark 7.2 (Computational verification). The companion dataset [Pap26] contains a sweep verifying the formula $\#\mathcal{A}_{\text{actual}}^{\text{cov}}/\#\mathcal{A}_{\text{expected}}^{\text{cov}} = 1.000$ exactly on 11 open-stratum operators.

Remark 7.3 (Beyond the open stratum). Outside (OS1)–(OS3), four classes of corrections occur: on the discriminant locus, resonance / logarithmic Stokes data appear; on the equal-modulus locus, the rotation in Theorem 3.3 becomes a rigid permutation up to shared rotation; on the zero-root locus, the Newton polygon itself wall-crosses; on the ω -resonance locus, anti-Stokes rays cluster beyond what (OS3) permits.

7.3 Nested ramification on wild data

Theorem 7.4 (Nested ramification: $\mu_{q'}$ -action on wild data). *Suppose the hypotheses of Theorem 4.6 hold: $\chi_w(w) = \chi_v(w^{q'})$ for some $q' \geq 2$, with χ_v admitting no further inner factorisation. Let $\text{MW}(L)$ denote the wild character variety of L at $x = \infty$. Then:*

1. *The minimal Levelt–Turrittin ramification cover for $\hat{L}$ is $x = t^q$ (not $x = \tau^{qq'}$): on the q -cover, the formal connection splits into rank-one summands with irregular types $Q_i(t) = c_i t^p + (\text{lower})$, and the c_i are exactly the elements of $\mathcal{C}(L)$ listed by Theorem 4.6.*
2. *The nested factorisation induces an additional $\mu_{q'}$ -action on the constellation: any $\eta \in \mu_{qq'}$ with $\eta^q = \zeta_{q'}$ acts by $c \mapsto \eta c$, preserving each v -fibre $\{c : c^{qq'} = v\}$ and permuting the q' choices of $w = c^q$ within that fibre. Combined with the cover-action $c \mapsto \zeta_q c$, the visible symmetry is $\mu_{\text{lcm}(q, q')} \subseteq \mu_{qq'}$, with equality iff $\gcd(q, q') = 1$.*
3. *The locus of irregular types $\{Q_i(t)\}_i$ compatible with the factorisation $\chi_w = \chi_v \circ (\cdot)^{q'}$ is $\mu_{q'}$ -invariant in the discrete part $\text{MW}(L)^{\text{disc}}$; it organises into a $\mu_{q'}$ -equivariant stratum.*

Proof sketch. Part (1). The Levelt–Turrittin ramification index of a single-slope scalar formal connection is the denominator q of its slope in lowest terms. The hypothesis $\chi_w(w) = \chi_v(w^{q'})$ constrains the multiset of leading coefficients $\{c_i\}$ but does not change the slope p/q of $N(L)$. The minimal cover therefore remains $x = t^q$, as in Theorems 3.1 and 7.1.

Part (2). By Theorem 4.6, $\mathcal{C}(L) = \bigsqcup_v \{c : c^{qq'} = v\}$; each v -fibre carries a free $\mu_{qq'}$ -action. The μ_q -action sits inside $\mu_{qq'}$ as $\{\zeta_{qq'}^{q'k}\}$, and the $\mu_{q'}$ -action of the nested factorisation lifts to a c -level action by $\eta \in \mu_{qq'}$ with $\eta^q = \zeta_{q'}$. The subgroup generated by both is $\mu_{\text{lcm}(q, q')}$.

Part (3). The discrete part $\text{MW}(L)^{\text{disc}}$ parametrises tuples $(Q_1, \dots, Q_N)$ of irregular-type polynomials modulo formal gauge and formal monodromy. The $\mu_{q'}$ -action of Part (2) induces a permutation $Q_i \mapsto Q_{\sigma(i)}$; the locus compatible with a fixed χ_v is therefore $\mu_{q'}$ -invariant. We assert equivariance only; the further refinement to a $\mu_{q'}$ -quotient of an un-symmetrised moduli space lies in the wild-character-variety formalism (Boalch [Boa14, Boa16]; Sabbah [Sab05]). $\square$

Remark 7.5 (Correction of a Phase-IV draft statement). An earlier Phase-IV draft claimed that the minimal Levelt–Turrittin cover for a nested-ramification operator becomes $x = \tau^{qq'}$, which would double-count anti-Stokes rays per pair. The corrected statement (Theorem 7.4(1)) holds the cover at $x = t^q$: nested factorisation is a refinement of the constellation and of the wild character variety stratification, not of the formal ramification index.

7.4 Multi-edge slope filtration on the anti-Stokes skeleton

Theorem 7.6 (Multi-edge slope filtration of Stokes data). *Suppose L has multi-edge Newton polygon as in Theorem 4.3, with edges E_α of slope $-p_\alpha/q_\alpha$ and per-edge constellations $\mathcal{C}_\alpha$ of size $N_\alpha = q_\alpha r_\alpha$, satisfying (OS1)–(OS5). Let $d = \text{lcm}(q_1, \dots, q_m)$ and consider the common cover $x = t^d$, with $P_\alpha := p_\alpha d/q_\alpha$. Then:*

1. *(Within-edge rays.) Pairs (i, j) both in $\mathcal{C}_\alpha$ contribute $2P_\alpha \binom{N_\alpha}{2}$ rays (with multiplicity) at t -angle spacing π/P_α .*

2. (Cross-edge rays.) For pairs $(i \in \mathcal{C}_\alpha, j \in \mathcal{C}_\beta)$ with $\alpha \neq \beta$, $P_\beta > P_\alpha$ (WLOG), the leading term of $Q_j^{(\beta)} - Q_i^{(\alpha)}$ on $x = t^d$ is $c_j t^{P_\beta}$ (using (OS5): $c_j \neq 0$), giving the anti-Stokes condition $\arg c_j + P_\beta \phi \equiv \pi/2 \pmod{\pi}$ and contributing $2P_\beta$ rays per cross-edge pair.
3. (Slope filtration.) The Stokes data of L decompose along the slope filtration $\hat{L} \simeq \bigoplus_\alpha L_\alpha$: Stokes constants within L_α live at slope $-p_\alpha/q_\alpha$; cross-edge Stokes constants ($L_\alpha \rightarrow L_\beta$ with $\alpha < \beta$) are organised along rays of the higher slope $-p_\beta/q_\beta$.

Proof sketch. Step 1 (Within-edge). By Theorem 4.3, $\hat{L} \simeq \bigoplus L_\alpha$ formally, and the Stokes structure of a direct sum is the direct sum of per-summand Stokes structures. Theorem 7.1 on the q_α -cover gives $2p_\alpha \binom{N_\alpha}{2}$ rays; lifting to the common cover via $s = t^{d/q_\alpha}$ multiplies each ray's multiplicity by d/q_α , yielding $2P_\alpha \binom{N_\alpha}{2}$ rays at spacing π/P_α .

Step 2 (Cross-edge). On the common cover, E_α 's summand has $Q_i^{(\alpha)}(t) = c_i t^{P_\alpha}$ and E_β 's has $Q_j^{(\beta)}(t) = c_j t^{P_\beta}$. Since slopes are distinct, $P_\alpha \neq P_\beta$; with $P_\beta > P_\alpha$ and (OS5), $Q_j^{(\beta)} - Q_i^{(\alpha)} = c_j t^{P_\beta} - c_i t^{P_\alpha} + (\text{lower})$ has leading term $c_j t^{P_\beta}$. The anti-Stokes analysis of Theorem 7.1 applied to this leading term gives $2P_\beta$ rays per pair.

Step 3 (Slope filtration). The Stokes-graded structure of a slope-filtered irregular connection is standard; see Sabbah [Sab05, §11], Boalch [Boa16, §3]. $\square$

7.5 p -visibility

The slope numerator p is invisible to the constellation (Theorem 4.8: $\mathcal{C}(L)$ depends on χ_w , not on p), but visible to the analytic anti-Stokes skeleton.

Theorem 7.7 (p -visibility on the cover). *Fix q and r and let χ_w be a fixed inner polynomial of degree r . For each $p \in \{p' : 1 \leq p' < q, \gcd(p', q) = 1\}$, let L_p denote a scalar operator whose Newton polygon has a single dominant edge of slope $-p/q$ with the prescribed χ_w . Then:*

1. (Constellation p -invariance.) $\mathcal{C}(L_p)$ is independent of p as a multiset, and the Δ -modulus partition $\{|\Delta_{ij}|\}$ is p -independent.
2. (p -visibility on the cover.) The anti-Stokes ray set $\mathcal{A}_p^{\text{cov}}$ on the cover $x = t^q$ depends on p : rays satisfy $\arg \Delta_{ij} + p\phi \equiv \pi/2 \pmod{\pi}$, so varying p rotates and dilates the ray pattern.
3. (Quantitative formula on the open stratum.) For χ_w generic with r distinct nonzero roots of pairwise distinct moduli and $q \geq 3$,

$$\#\mathcal{A}_p^{\text{cov}} = 2pq \left(r + q \binom{r}{2} \right).$$

For $q = 2$, $\#\mathcal{A}_p^{\text{cov}} = 2p(r + 4 \binom{r}{2})$.

Hence Phase-III's constellation-level p -invariance (P1) holds, while the analytic anti-Stokes skeleton refines it via the p -proportional ray count.

Proof. Part (1). $\chi(c) = \chi_w(c^q)$ depends only on χ_w , so $\mathcal{C}(L_p) = \text{Roots}(\chi)$ is independent of p .

Part (2). Theorem 7.1 gives $\arg \Delta_{ij} + p\phi \equiv \pi/2 \pmod{\pi}$; solving for ϕ shows the spacing π/p between adjacent rays of a fixed pair, manifestly p -dependent.

Part (3) (Counting). With $\mathcal{C}(L) = \bigsqcup_{\alpha=1}^r R_\alpha$, $R_\alpha = \{\zeta_q^j z_\alpha\}$:

- *Within-ring.* For each ring, the $\binom{q}{2}$ unordered pairs $\{j, k\}$ give $\Delta_{ij} = z_\alpha \zeta_q^k (\zeta_q^{j-k} - 1)$, whose argument modulo π depends only on $(j+k) \pmod{q}$, taking q distinct values. Each direction contributes $2p$ rays, yielding $2pq$ per ring and $2pqr$ total.
- *Cross-ring.* The $q^2 \binom{r}{2}$ unordered cross-ring pairs generically have pairwise distinct $\arg \Delta_{ij} \pmod{\pi/p}$ (open stratum), each contributing $2p$ rays for $2p \cdot q^2 \binom{r}{2}$ total.

[x = 1.0cm, y = 1.0cm, font = , atlas/.style = draw, rounded corners=2pt, minimum width=1.55cm, minimum height=0.55cm, align=center, fill=white, inner sep=1.5pt, deg/.style = draw, rounded corners=2pt, minimum width=1.85cm, minimum height=0.55cm, align=center, fill=gray!8, inner sep=1.5pt, ref/.style = draw, dashed, rounded corners=2pt, minimum width=2.05cm, minimum height=0.55cm, align=center, fill=gray!4, inner sep=1.5pt, grp/.style = draw, rounded corners=4pt, inner sep=6pt, fill=gray!2, very thin, conn/.style = -Latex[length=2.0mm,width=1.4mm], very thin, gray]

[font=] at (0, 4.7) $-1/2$; [font=] at (2, 4.7) $-1/3$; [font=] at (4, 4.7) $-1/4$; [font=] at (6, 4.7) $-2/3$; [font=] at (8, 4.7) $-3/2$;

[font=] at (-1.4, 3.6) $r = 1$; [font=] at (-1.4, 2.6) $r = 2$; [font=] at (-1.4, 1.6) $r = 3$;

[atlas] (a11) at (0, 3.6) $(1/2, 1)$; [atlas] (a12) at (2, 3.6) $(1/3, 1)$; [atlas] (a13) at (4, 3.6) $(1/4, 1)$; [atlas] (a14) at (6, 3.6) $(2/3, 1)$; [atlas] (a15) at (8, 3.6) $(3/2, 1)$;

[atlas] (a21) at (0, 2.6) $(1/2, 2)$; [atlas] (a22) at (2, 2.6) $(1/3, 2)$; [atlas] (a23) at (4, 2.6) $(1/4, 2)$; [atlas] (a24) at (6, 2.6) $(2/3, 2)$; [atlas] (a25) at (8, 2.6) $(3/2, 2)$;

[atlas] (a31) at (0, 1.6) $(1/2, 3)$; [atlas] (a32) at (2, 1.6) $(1/3, 3)$; [atlas] (a33) at (4, 1.6) $(1/4, 3)$; [atlas] (a34) at (6, 1.6) $(2/3, 3)$; [atlas] (a35) at (8, 1.6) $(3/2, 3)$;

[deg] (d1) at (1.0, 0.2) equal-mod; [deg] (d2) at (3.2, 0.2) double root; [deg] (d3) at (5.4, 0.2) zero root; [deg] (d4) at (7.6, 0.2) accident. sym.;

[grp, fit = (a11) (a35) (d1) (d4), label=[font=, anchor=south] *above:23 single-edge classes*] (G23) ;

[ref] (r1) at (1.5, -1.7) multi-edge
(M1: $\prod \chi^{(\alpha)}$); [ref] (r2) at (4.5, -1.7) nested ramif.
(N1: $\mu_{q'}$ -pullback); [ref] (r3) at (7.5, -1.7) prime- q arith.
(D2: Kummer dichotomy);

[grp, fit = (r1) (r3), label=[font=, anchor=north] *below:+16 refining strata $\Rightarrow$ 39 classes total*] (G39) ;

[conn] (a31) – (r1.north west); [conn] (a23) – (r2.north); [conn] (a35) – (r3.north east);

[font=, align=left, anchor=west] at (10.0, 2.4) $F : L \mapsto C(L)$

refinement
 $\downarrow$
 $F^{\text{ref}};$

Figure 1: The F-image stratification: 23 single-edge constellation classes, refining to 39 once multi-edge polygons, nested ramification, and prime- q arithmetic strata are tracked (Theorems 4.3, 4.6 and 5.1). The 23-class atlas is the slope-by-rank grid plus four named degenerate strata (equal-modulus, double root, zero root, accidental symmetry); the +16 refining strata come from the multi-edge product (M1), nested ramification (N1), and the prime- q Kummer dichotomy (D2).

Summing: $\#\mathcal{A}_p^{\text{cov}} = 2pqr + 2pq^2\binom{r}{2} = 2pq(r + q\binom{r}{2})$. For $q = 2$, the within-ring direction count is $q^2 = 4$ (no overcounting factor), giving $\#\mathcal{A}_p^{\text{cov}} = 2p(r + 4\binom{r}{2})$. $\square$

Remark 7.8 (Phase-IV verification). On the PINV family with $q = 5$, $r = 2$, $N = 10$, the formula gives $\#\mathcal{A}_p^{\text{cov}} = 2p \cdot 5 \cdot (2 + 5) = 70p$. The Phase-IV measurement [Pap26] records 70, 140, 210, 280 for $p = 1, 2, 3, 4$, exact agreement with $70p$.

Remark 7.9 (Conceptual content). Theorem 7.7 resolves an apparent tension between P1 (the constellation does not see p) and the analytic Stokes geometry (which manifestly depends on p via the t^p exponent in Q_i). The discrete and analytic invariants are not in contradiction: the constellation coarsens away p , and the anti-Stokes skeleton recovers it via the ray spacing.

8 The 23-class atlas and the constellation zoo

8.1 Indexing the image of F

By Theorem 3.1 and Theorem 3.2, an element of the image of F is determined, up to global rotation, by the triple (q, r, π) where π is the modulus-with-multiplicity partition of the roots of χ_w . The slope numerator p is invisible at this level (Theorem 7.7(1)), so the atlas is indexed by (q, r, π) alone.

8.2 Taxonomy diagram

The taxonomy diagram of Figure 1 encodes the image stratification visually. A precise textual specification (axes, sub-tile structure, encoding conventions, degeneracy island, legend) is recorded

WKB Exponent Constellations — Representative Classes + Stress Tests

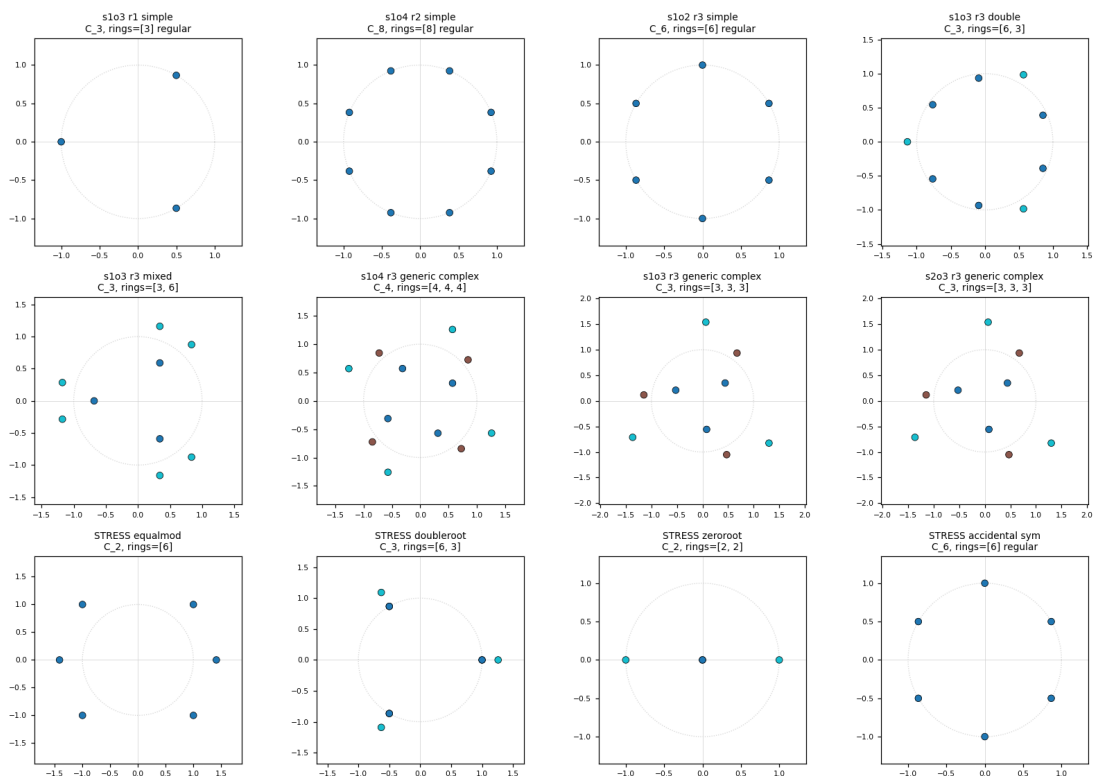


Figure 2: Representative constellations across the F-image. The 3×4 grid covers the principal strata. Top row (A1–A4): simple / binomial classes — single regular polygon (a single Galois orbit of χ_w). Middle row (B1–B4): double and generic-complex classes — two or three concentric μ_q -orbits. Bottom row (C1–C4): the four named degenerate strata. Panels A3 and C4 are visually identical 6-gons but live in different cells: A3 at slope $1/3$ has minimum symmetry C_3 enhanced to C_6 ; C4 at slope $1/2$ has C_2 enhanced to C_6 . The constellation alone does not recover the cell — the Newton-polygon denominator q is required. This is the visible signature that F is many-to-one and that accidental symmetry is a feature of every simple-class operator. The full dataset of plots is in the companion repository [Pap26].

in the companion dataset [Pap26].

8.3 Constellation zoo

Figure 2 shows twelve representative panels. The full enumeration of all 23 classes, with explicit operator representatives, χ_w data, and panel pointers, lives in the companion dataset [Pap26].

9 Related work

ADE singularities and discrete classification. As a methodological precedent we take the ADE classification of simple singularities (Arnol’d [Arn76], Brieskorn [Bri71]) and of finite subgroups of $SU(2)$ (McKay [McK80]): a small discrete invariant—a Dynkin diagram—organises an apparently heterogeneous zoo of analytic objects, with strata governed by simple-root combinatorics and degenerations corresponding to root-system collisions. Our 23 constellation classes occupy an analogous slot: a finite combinatorial atlas indexed by the discrete data (q, r, π, χ_w) , with degenerate strata produced by collisions among the inner roots w_i (the analogue

of root coincidences). Unlike the ADE setting, our classes are not labelled by Dynkin diagrams—the relevant combinatorics is the partition of χ_w ’s roots by modulus and Galois orbit—but the *style* of classification (discrete invariants, stratified moduli, degeneration loci) is closely modelled on it.

Formal classification: Levelt–Turrittin and descendants. The Levelt–Turrittin theorem (Levelt [Lev75], Turrittin [Tur55]; modern accounts in Sabbah [Sab07] and Sabbah–Mochizuki [Moc11]) gives a formal classification of meromorphic connections by *irregular type*: a multiset of exponential factors $e^{P_i(x^{1/q})}$ together with regular-singular twists. Birkhoff–Trjitzinsky [BT33] and the q -difference theories of Sauloy [Sau03], Ramis [Ram97, RSZ13], and Zhang [Zha98] play the parallel role for difference and q -difference operators. Each of these theories classifies *the operator* up to formal or analytic equivalence; our object is different. We classify the *image* of the WKB-extraction functor F , i.e. the discrete geometry that $N(L)$ and its leading-edge coefficients deposit on the complex plane. The connection is one-way: the Levelt–Turrittin/Birkhoff–Trjitzinsky exponents are recovered from $\mathcal{C}(L)$ together with auxiliary phase data, but $\mathcal{C}(L)$ itself is coarser and admits a clean independent classification.

Stokes graphs and spectral networks. A third neighbour is the Stokes-graph classification of Painlevé and other low-rank equations (Kawai–Takei [KT05], Iwaki–Nakanishi [IN14], Aoki–Kawai–Takei [AKT01]), and more broadly the spectral-network programme of Gaiotto–Moore–Neitzke [GMN13]. There, classification proceeds via the *combinatorial topology of Stokes lines* in the complex plane, which depends on $c_i - c_j$ rather than $|c_i|$. Our ring-modulus partition is therefore *not* a Stokes invariant and does not reproduce spectral-network combinatorics; instead it captures the strictly weaker but exact information of the multiset $\{c_i^q\}$. This makes F a coarsening of the spectral-network functor, which we view as an asset: the coarser invariant is computable directly from Newton-polygon data and admits a closed-form classification, whereas Stokes graphs in general do not.

Wild character varieties and irregular Riemann–Hilbert. The geometry of *wild character varieties* and the irregular Riemann–Hilbert correspondence (Boalch [Boa14, Boa16], Sabbah [Sab05], Mochizuki [Moc11]) produces moduli stacks parametrising local data of irregular connections; our F -image is a discrete shadow of such a stack restricted to scalar operators with one dominant edge. We do not attempt to lift the present classification to the level of stacks or to incorporate Stokes structures intrinsically; rather, we exhibit a self-contained classification of the discrete part of the local data, with the analytic refinement of §7 attaching the first layer of Stokes information (anti-Stokes directions; slope-graded Stokes filtration) but stopping short of explicit Stokes constants. The contribution is best understood as building a separately classifiable layer of the local picture—a Newton-polygon WKB stratum—on which the fuller analytic theories can subsequently be erected.

10 Discussion and future directions

The natural object of study, in our view, is not any individual operator but the constellation functor F itself: the map from locally-irregular linear models to their WKB exponent multisets. Once F is named, the questions become geometric—what is its image, what are its fibres, what are its strata, how does it deform—and admit answers in closed form when restricted to one dominant Newton edge. The 23 classes catalogued here populate the lowest-dimensional cells of $\text{Im}(F)$; their adjacencies trace out the discriminant, equal-modulus, and zero-root walls along which the structure changes.

The natural sequel is to walk the same programme up the complexity ladder. Theorems 4.3, 4.6 and 7.6 already step one rung: multi-edge Newton polygons (where F acquires a slope-

filtration target) and nested ramification (where the visible symmetry inflates from μ_q to $\mu_{\text{lcm}(q,q')}$). Beyond:

- **Coupled systems.** Matrix and meromorphic-connection systems have root systems on each edge; their WKB target is a wild character variety. Theorem 6.1 treats only the block-diagonal case. Genuinely coupled systems (non-semisimple leading symbol) lie outside the present classification and connect naturally to Boalch’s wild non-abelian Hodge theory [Boa16].
- **Stokes constants and the analytic refinement.** The anti-Stokes skeleton of Theorems 7.1, 7.6 and 7.7 attaches *directions* to constellations; the next layer is the attachment of *Stokes constants* along each ray. The locus of Stokes-equivalent operators above a fixed constellation is the kernel of a refined functor $\tilde{F}$ whose image is a wild-character-variety stratum.
- **Difference and q -difference Stokes lifts.** The analytic theorems are stated for $\square = D$; the analogues for S (Birkhoff–Trjitzinsky) and T (Ramis θ -Stokes) require separate analytic foundations and are deferred. The constellation identities (Theorems 3.1, 4.3, 4.6, 5.1 and 6.1) hold verbatim across equation types.
- **Composite- q Kummer dichotomy.** Theorem 5.1 is sharp for prime q within the Phase-III search window. The composite- q version requires the refinement of Theorem 5.2 and the transcendental-degree statement of Theorem 5.3.

We expect the functorial reframing to be useful precisely because it makes a single, universal target geometry visible behind a great many local stories that, treated one at a time, have looked unrelated. The classifiable object is the functor; the discrete combinatorics of the constellation sits transversally to the analytic theory—coarser, computable, and structurally separable.

11 Reproducibility note

All computational artifacts—the operator atlas, characteristic polynomials, constellations, clusters, conjectures, counterexample catalogue, taxonomy specification, and rendered figures—are bundled in the companion Zenodo dataset [Pap26]. The dataset is organised under the directory structure

```
/code /data /figures /reports /theorems /paper-draft /fleets
```

with `README.md` at the root. The five fleet pipelines (`fleet.py`, `fleet3.py`, `fleet4.py`, `fleet5.py`) are deterministic and self-contained (Python 3.10+, `sympy`, `numpy`, `matplotlib`); no random seeds are used. Running each fleet in sequence reproduces the JSON atlas and the rendered constellation zoo (`constellations.png`).

The proofs of Theorems 4.3, 4.6, 5.1, 6.1, 7.1, 7.4, 7.6 and 7.7 are also bundled in the dataset as the file `proofs.md`, in long-form expository style with explicit cross-references to Phase-III/IV evidence. The integration map (`integration_map.md`) records, for each theorem, the dataset artifact that supplies its empirical evidence. The counterexample catalogue (`counterexamples.json`) records the strata on which naive forms of the theorems fail; each theorem statement above includes a remark guarding against the relevant counterexamples.

Citation. If you use this work, please cite both the paper and the dataset; the dataset DOI (placeholder 10.5281/zenodo.XXXXXXX) will be assigned by Zenodo on publication of v1.0.

Computational and AI Disclosure

Computational environment. All numerical and symbolic work in this paper was carried out in Python 3.10+ using `sympy` (exact symbolic algebra), `fractions` and `numpy` (exact rational and floating-point arithmetic), and `matplotlib` (figure rendering). No randomised algorithms were used and no random seeds were set: every JSON artifact in the companion dataset [Pap26] is reproduced bitwise on re-execution. The manuscript was typeset with MiKTeX/TeX Live using the package set listed in the preamble; the bibliography was processed with `bibtex` against `references.bib`.

Automated pipelines (Phases I–V). The computational atlas was produced by five fleet scripts, each implementing one phase of the investigation:

- `fleet.py` (Phases I/II) — generates the single-edge operator atlas, computes Newton polygons, extracts characteristic polynomials χ and inner polynomials χ_w , clusters the exponent constellations, and emits the initial conjecture list.
- `fleet3.py` (Phase III) — extends the atlas to multi-edge polygons, 2×2 and 3×3 systems, resonant slopes ($q \leq 7$), and nested ramification; computes Δ_{ij} partitions, Galois-orbit invariants, and the inner-polynomial $\mathbb{Z}$ -resonance dichotomy.
- `fleet4.py` (Phase IV) — lifts the discrete invariants to the analytic layer via the Levelt–Turrittin decomposition, predicts anti-Stokes rays from the Δ_{ij} geometry, and cross-checks the predictions against numerical Stokes computations.
- `fleet5.py` (Phase V) — consolidates the conjectures into theorem–proof pairs, runs each statement against the counterexample catalogue, and emits the proof skeletons recorded in `proofs.md` with placements specified by `integration_map.md`.

The pipelines are deterministic and self-contained; running them in sequence reproduces every JSON file, the rendered constellation zoo (`constellations.png`), and the proof skeletons.

Role of AI assistance. This work was conducted as independent research with AI-assisted execution (GitHub Copilot CLI driving an Anthropic Claude Opus 4.7 model). The assistant was used in five clearly demarcated roles:

- (i) *Conjecture selection.* The assistant scanned the Phase-I–III invariant tables for candidate universality laws; the author selected which conjectures to promote to the main text.
- (ii) *Proof-skeleton generation.* For each promoted conjecture the assistant produced a draft proof skeleton naming the algebraic ingredients (Newton-polygon combinatorics, μ_q -pullback, χ_w factorisation, Levelt–Turrittin, determinant polygon); the author verified each step and rewrote the load-bearing arguments.
- (iii) *Counterexample guarding.* The assistant cross-referenced each candidate theorem against `counterexamples.json` and proposed genericity hypotheses (prime q , single-edge, non-resonant inner polynomial) that survive the catalogued exceptions; the author adjudicated which restrictions were accepted into the final statements.
- (iv) *Refinement passes.* The assistant unified notation, fixed cross-references, and polished prose; all theorem statements, remarks, and limitations were reviewed by the author.
- (v) *Manuscript assembly.* The assistant inserted the theorem–proof blocks into the section placements recorded in `integration_map.md`, assembled the bibliography, and produced the arXiv and journal submission bundles.

No proprietary or closed-source models were used beyond the publicly-available AI assistant named above; no model was fine-tuned on the project’s data, and no AI-mediated literature summaries were used in place of primary sources for citations.

Human accountability. All mathematical statements, theorem formulations, proofs, and classification claims in this paper were reviewed, validated, and finalised by the author. In particular, the structural law of Theorem 3.1, the multi-edge and nested-ramification factorisations of Theorems 4.3 and 4.6, the prime- q Kummer dichotomy of Theorem 5.1, the system pullback of Theorem 6.1, the four analytic-lifting theorems Theorems 7.1, 7.4, 7.6 and 7.7, and the 23-class taxonomy of Section 8 reflect human mathematical judgment. The author retains sole accountability for all claims, methodology choices, and the published result.

Reproducibility. The complete computational record — operators, Newton polygons, characteristic polynomials, constellations, clusters, counterexamples, Phase-III/IV invariants, proof skeletons, integration map, and rendered figures — is deposited as the companion Zenodo dataset [Pap26], licensed CC-BY-4.0 (text and data) and MIT (code and pipelines). All code, fleet scripts, and sample artifacts are publicly available at <https://github.com/papanokechi/wkb-newton-polygons>, which mirrors the Zenodo v2.0 dataset and provides a stable, versioned record of the computational workflow. The dataset DOI is 10.5281/zenodo.XXXXXXX (placeholder, assigned by Zenodo at publication of v1.0). See Section 11 for the command sequence that reproduces every artifact.

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